

Volatility Modeling and Risk Forecasting with GARCH

1. Introduction

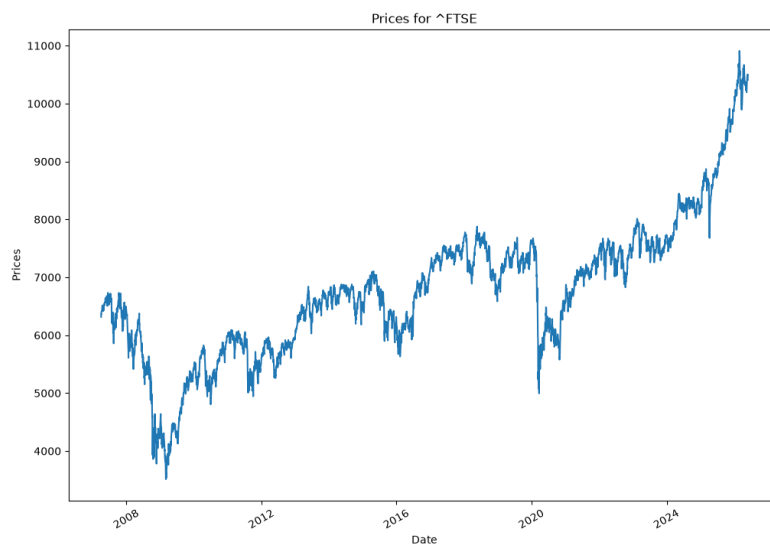
This project analyzes volatility in S&P 500, Euro Stoxx 50, Nikkei 225, and FTSE 100. It uses daily closing prices from Yahoo Finance to compute log returns, examine volatility clustering through EDA, and perform preliminary analysis that motivates GARCH modeling. The project then estimates AR(1)-GARCH(1,1) models with Student-t innovations to model conditional volatility, generate volatility forecasts, and evaluate 99% Value-at-Risk performance through an out-of-sample backtest.

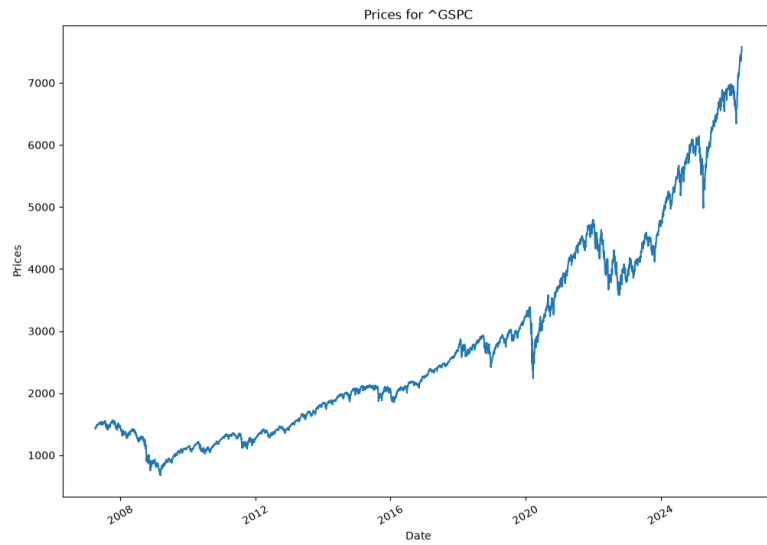
2. Data

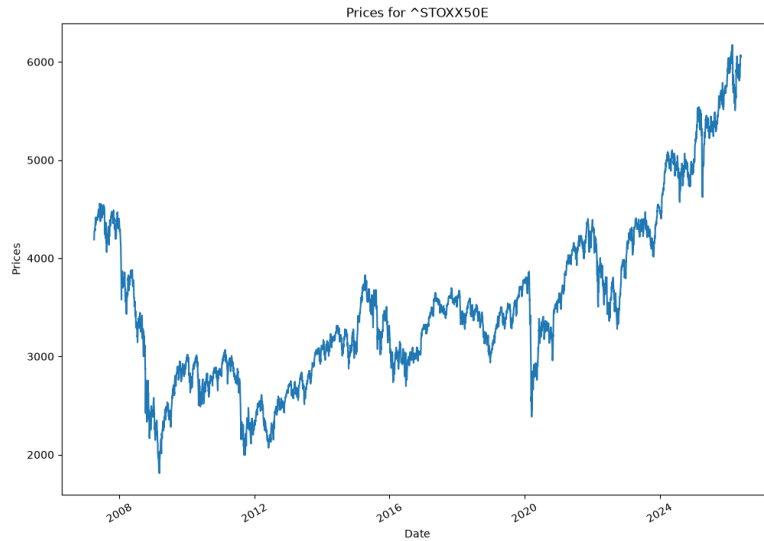
The data are obtained from Yahoo Finance using the yfinance package and include the daily closing prices of S&P 500, Euro Stoxx 50, Nikkei 225, and FTSE 100 between 2007-04-01 and 2026-04-01.

3. Exploratory Data Analysis

3.1 Price Series

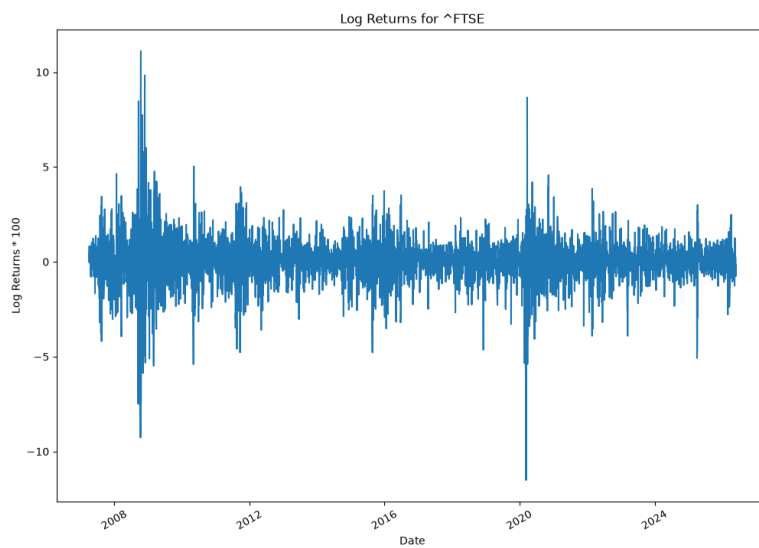


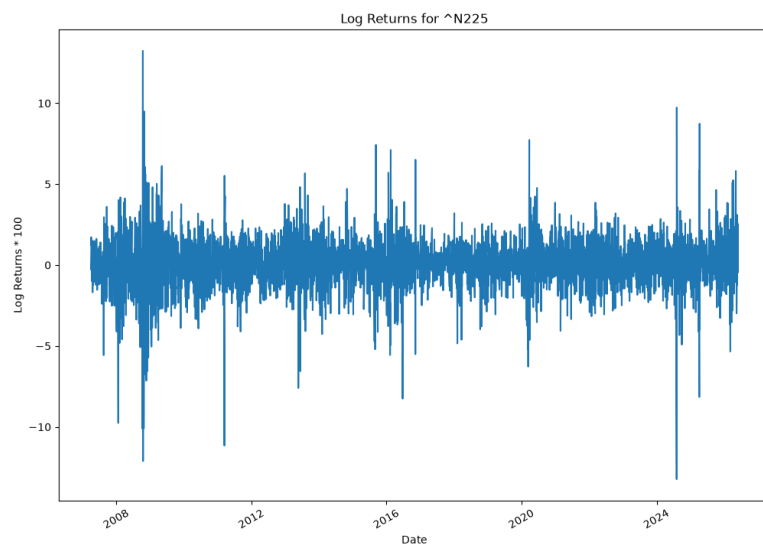
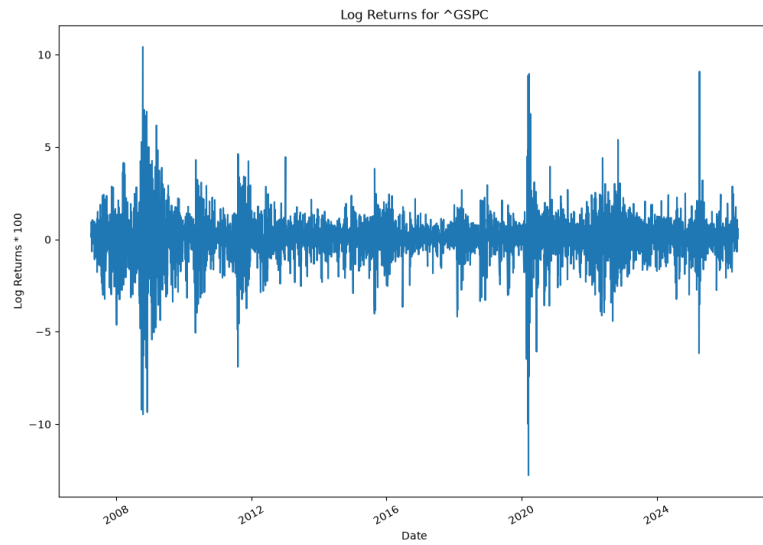


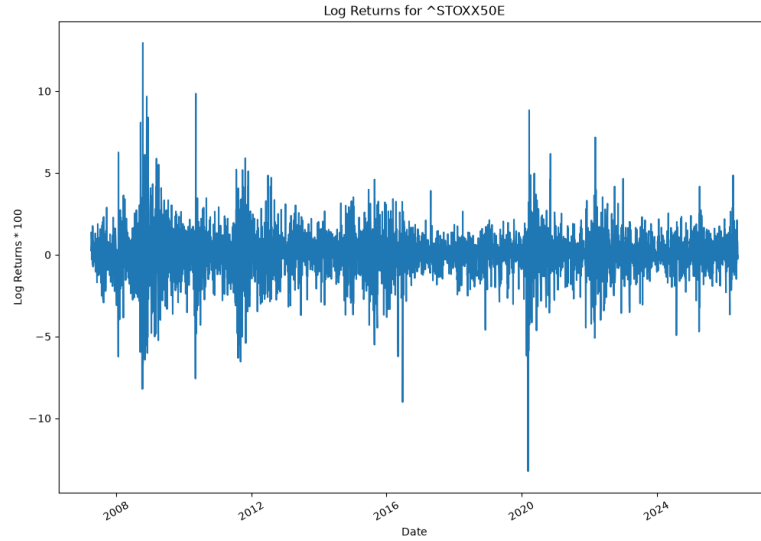


3.2 Return Series

The log return plots show that returns fluctuate around zero but exhibit clear periods of large clustered shocks (e.g. 2008 financial crisis and 2020 COVID-19 pandemic).



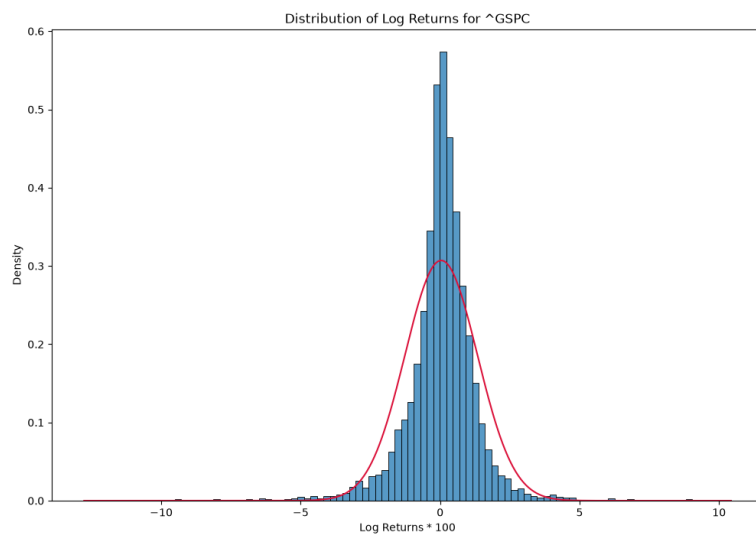
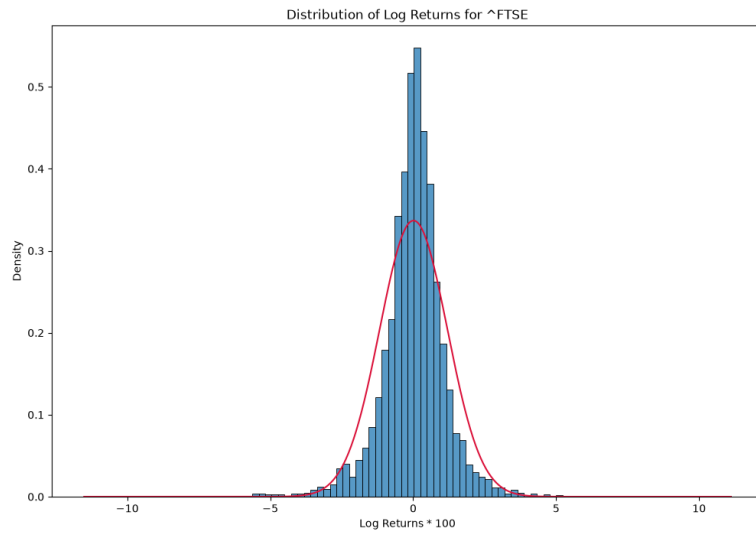


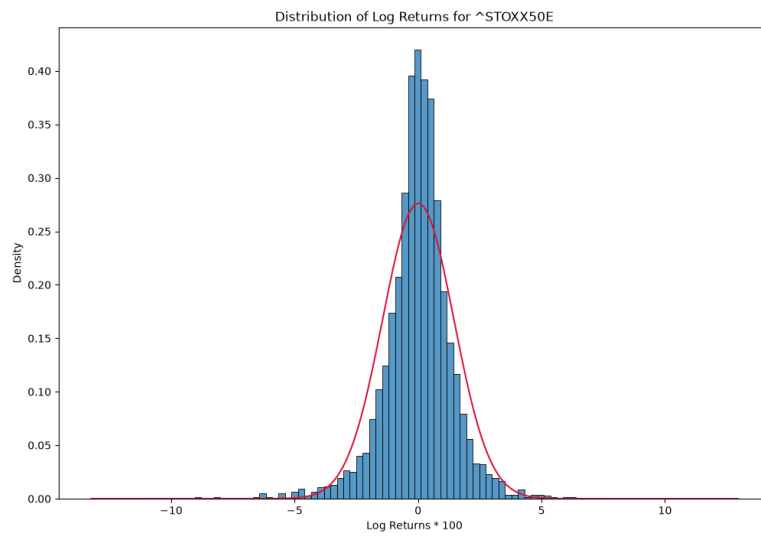
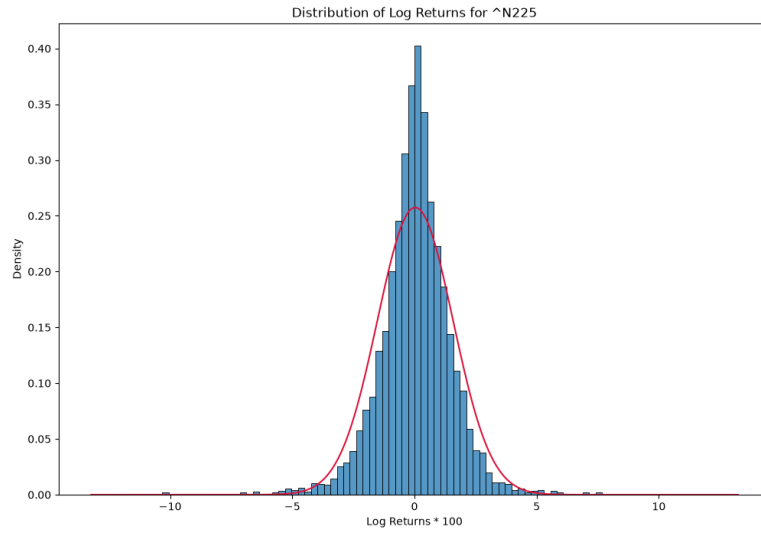


3.3 Return Distributions

The return distributions are centered close to zero but show negative skewness and high kurtosis, indicating heavier tails than a normal distribution.

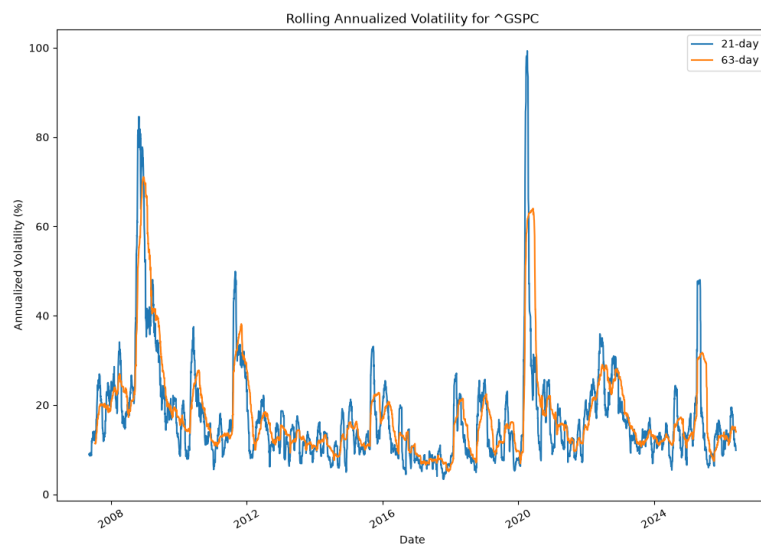
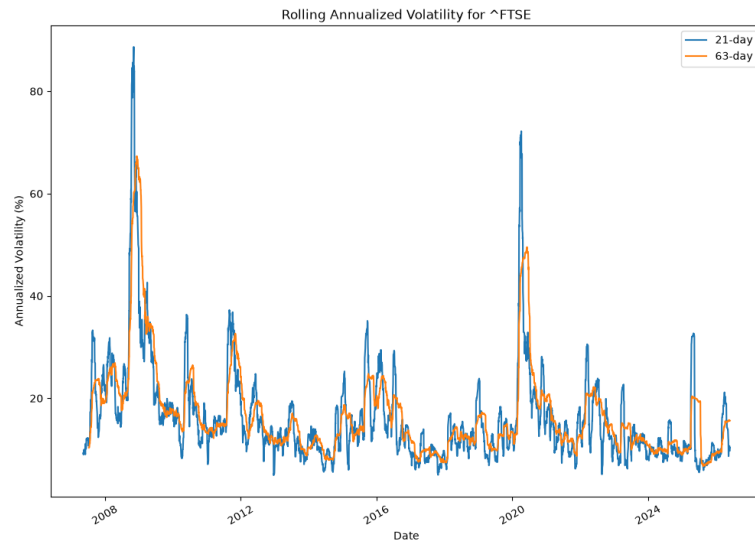
Index	Mean	Std. Dev.	Skewness	Kurtosis
FTSE	0.0114	1.1844	-0.2994	11.1410
GSPC	0.0382	1.2994	-0.6253	11.1280
N225	0.0311	1.5496	-0.4663	8.3043
STOXX50E	0.0084	1.4447	-0.2565	8.0194

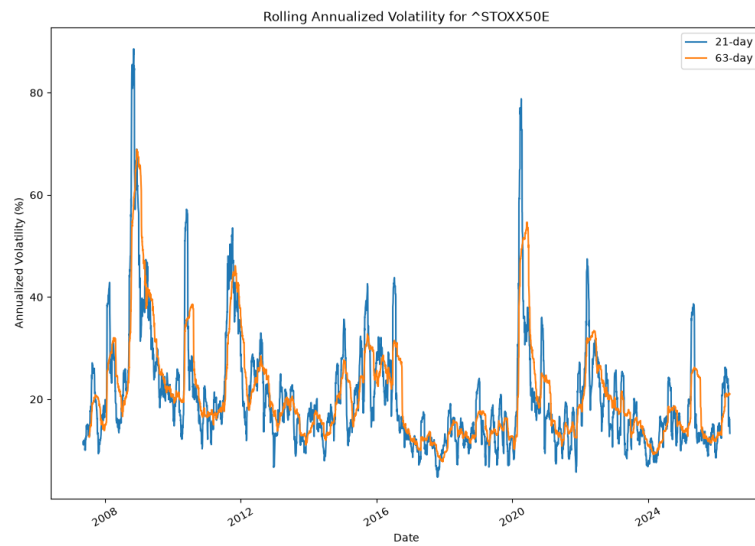
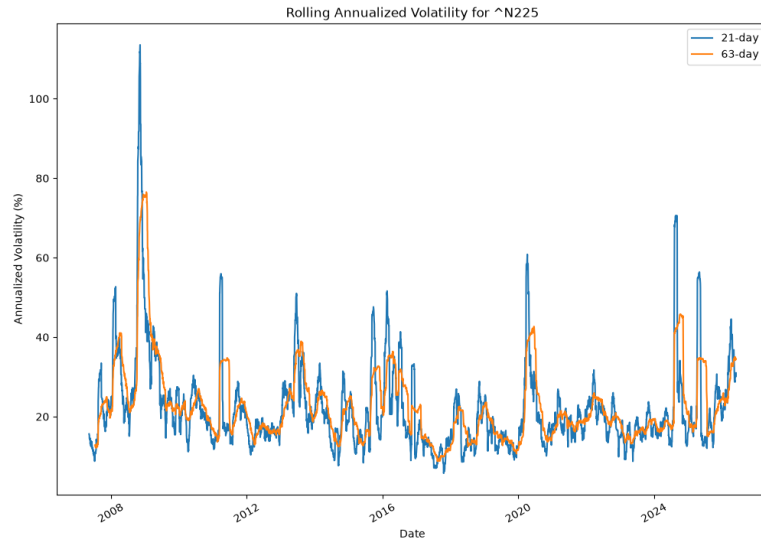




3.4 Rolling Volatility

The rolling volatility plots once again show that volatility changes over time and tends to remain elevated after large market shocks.





4. Preliminary Tests

4.1 Ljung-Box Tests on Log Returns

The p-value of the Ljung-Box Tests on squared log returns is very small (numerically zero), which indicates volatility clustering and suggests GARCH modeling.

index	series	lag	lb_stat	lb_pvalue
^FTSE	returns	10	47.7769	6.817e-07
^FTSE	returns	20	63.5542	1.979e-06
^FTSE	squared_returns	10	2813	0
^FTSE	squared_returns	20	4199	0
^GSPC	returns	10	137.7233	1.233e-24
^GSPC	returns	20	184.5889	1.224e-28
^GSPC	squared_returns	10	5061	0
^GSPC	squared_returns	20	6666	0
^N225	returns	10	25.1781	0.005
^N225	returns	20	51.4986	0.0001347
^N225	squared_returns	10	2012	0
^N225	squared_returns	20	2642	0
^STOXX50E	returns	10	34.4138	0.0001571
^STOXX50E	returns	20	45.2548	0.001
^STOXX50E	squared_returns	10	1690	0
^STOXX50E	squared_returns	20	2409	0

4.2 Stationarity Tests

The ADF tests reject the unit-root null for time series of all four indices, so the stationarity condition is satisfied for GARCH modeling.

index	ADF Statistic	p-value	lags used	observations	1% critical value	5% critical value	10% critical value
^FTSE	-28.0801	0	5	4370	-3.4318	-2.8622	-2.5671
^GSPC	-15.6464	1.617e-28	17	4358	-3.4319	-2.8622	-2.5671
^N225	-70.3178	0	0	4375	-3.4318	-2.8622	-2.5671
^STOXX50E	-31.4142	0	4	4371	-3.4318	-2.8622	-2.5671

4.3 ARCH LM Tests

The ARCH LM tests show very strong evidence of ARCH effects in all four return series after fitting an AR(1) mean model. This suggests that a GARCH-type model is appropriate for modeling these series.

index	mean_model	lags	lm_stat	lm_pvalue	f_stat	f_pvalue
^FTSE	AR(1)	10	961.7744	3.198e-200	123.0391	1.086e-226
^GSPC	AR(1)	10	1235	4.179e-259	171.7683	2.33e-305
^N225	AR(1)	10	785.6011	2.569e-162	95.5562	2.518e-179
^STOXX50E	AR(1)	10	675.4491	1.168e-138	79.7057	6.234e-151

5. Model Specification

$$\begin{aligned}
r_t &= \mu + \phi r_{t-1} + \varepsilon_t, \\
\varepsilon_t &= \sigma_t z_t, \\
\sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \\
z_t &\sim t_\nu.
\end{aligned}$$

6. Estimation Results

6.1 Model Fit

index	model	log_likelihood	AIC	BIC
^FTSE	AR(1)-GARCH(1,1)-t	-5909	1.183e+04	1.187e+04
^GSPC	AR(1)-GARCH(1,1)-t	-6017	1.205e+04	1.208e+04
^N225	AR(1)-GARCH(1,1)-t	-7392	1.48e+04	1.483e+04
^STOXX50E	AR(1)-GARCH(1,1)-t	-6939	1.389e+04	1.393e+04

6.2 Parameter Estimates

For all four indices, the AR(1)-GARCH(1, 1) model shows highly significant volatility clustering with persistent GARCH effects and fat-tailed return shocks.

index	model	parameter	estimate	std_error	t_stat	p_value
^FTSE	AR(1)-GARCH(1,1)-t	Const	0.0521	0.0118	4.4105	1.031e-05
^FTSE	AR(1)-GARCH(1,1)-t	^FTSE[1]	-0.0056	0.0154	-0.3646	0.7154
^FTSE	AR(1)-GARCH(1,1)-t	omega	0.0311	0.0071	4.3921	1.123e-05

index	model	parameter	estimate	std_error	t_stat	p_value
^FTSE	AR(1)- GARCH(1,1)- t	alpha[1]	0.1311	0.0192	6.8379	8.037e-12
^FTSE	AR(1)- GARCH(1,1)- t	beta[1]	0.8485	0.021	40.459	0
^FTSE	AR(1)- GARCH(1,1)- t	nu	5.8568	0.5162	11.3469	7.686e-30
^GSPC	AR(1)- GARCH(1,1)- t	Const	0.1104	0.0113	9.7755	1.435e-22
^GSPC	AR(1)- GARCH(1,1)- t	^GSPC[1]	-0.0575	0.0147	-3.9107	9.202e-05
^GSPC	AR(1)- GARCH(1,1)- t	omega	0.021	0.0047	4.5121	6.42e-06
^GSPC	AR(1)- GARCH(1,1)- t	alpha[1]	0.1471	0.0151	9.7149	2.604e-22
^GSPC	AR(1)- GARCH(1,1)- t	beta[1]	0.8522	0.0132	64.4541	0
^GSPC	AR(1)- GARCH(1,1)- t	nu	4.9918	0.3798	13.1448	1.823e-39
^N225	AR(1)- GARCH(1,1)- t	Const	0.0834	0.0174	4.8091	1.516e-06
^N225	AR(1)- GARCH(1,1)- t	^N225[1]	-0.0392	0.0155	-2.5382	0.0111
^N225	AR(1)- GARCH(1,1)- t	omega	0.0645	0.0169	3.819	0.000134
^N225	AR(1)- GARCH(1,1)- t	alpha[1]	0.1144	0.0184	6.2288	4.7e-10
^N225	AR(1)- GARCH(1,1)- t	beta[1]	0.8597	0.022	39.0374	0

index	model	parameter	estimate	std_error	t_stat	p_value
$\hat{N}225$	AR(1)- GARCH(1,1)- t	nu	6.5603	0.6572	9.9818	1.831e-23
$\hat{STOXX50E}$	AR(1)- GARCH(1,1)- t	Const	0.0762	0.0153	4.9783	6.416e-07
$\hat{STOXX50E}$	AR(1)- GARCH(1,1)- t	$\hat{STOXX50E}$	0.1331	0.0149	-2.2292	0.0258
$\hat{STOXX50E}$	AR(1)- GARCH(1,1)- t	omega	0.0353	0.0103	3.4067	0.0006576
$\hat{STOXX50E}$	AR(1)- GARCH(1,1)- t	alpha[1]	0.1098	0.0192	5.7258	1.03e-08
$\hat{STOXX50E}$	AR(1)- GARCH(1,1)- t	beta[1]	0.8781	0.0206	42.5703	0
$\hat{STOXX50E}$	AR(1)- GARCH(1,1)- t	nu	5.3101	0.4259	12.4685	1.109e-35

6.3 Volatility Persistence

The large values of $\alpha[1] + \beta[1]$ suggest high volatility persistence across four indices, with $\hat{GSPC}$ showing near-integrated volatility persistence at 0.9993, followed by $\hat{STOXX50E}$, $\hat{FTSE}$, and $\hat{N225}$.

index	omega	alpha[1]	beta[1]	alpha_beta
$\hat{GSPC}$	0.021	0.1471	0.8522	0.9993
$\hat{STOXX50E}$	0.0353	0.1098	0.8781	0.988
$\hat{FTSE}$	0.0311	0.1311	0.8485	0.9795
$\hat{N225}$	0.0645	0.1144	0.8597	0.974

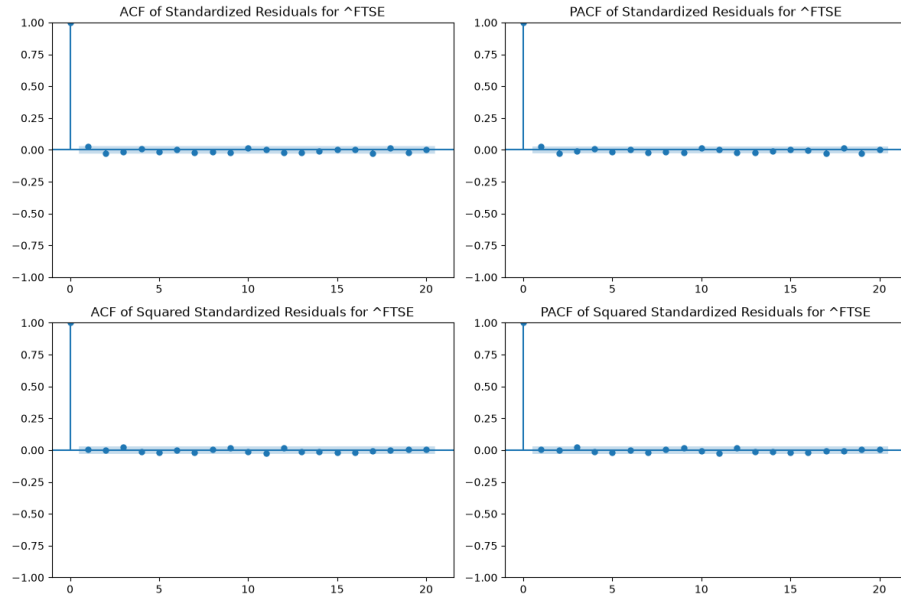
7. Model Diagnostics

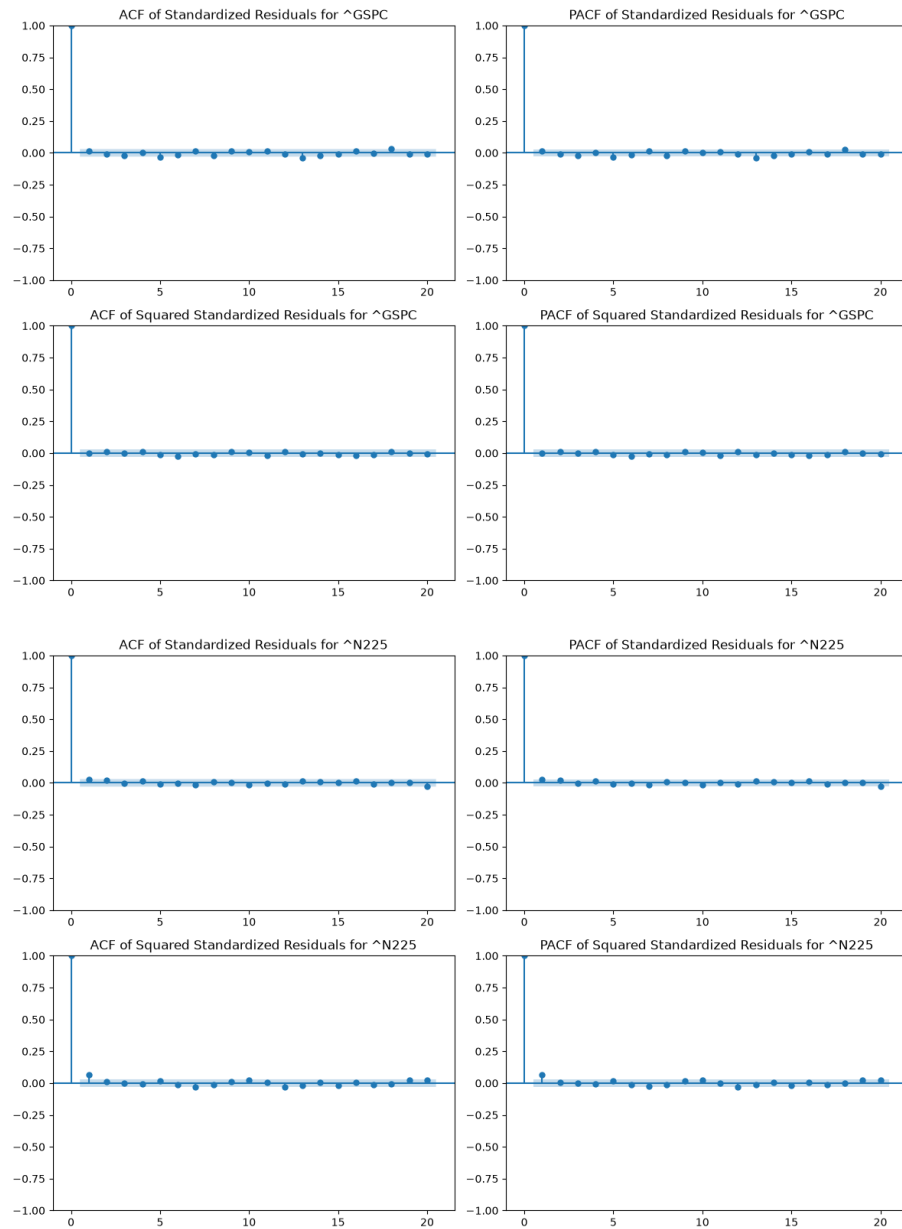
The Ljung-Box tests show that the AR(1)-GARCH(1, 1) model has sufficiently removed autocorrelation in the standardized residuals across four indices. For squared standardized residuals, the model performs well in removing autocorrelation in $\hat{FTSE}$ and $\hat{STOXX50E}$, but it does not fully capture volatility clustering for $\hat{GSPC}$ and $\hat{N225}$.

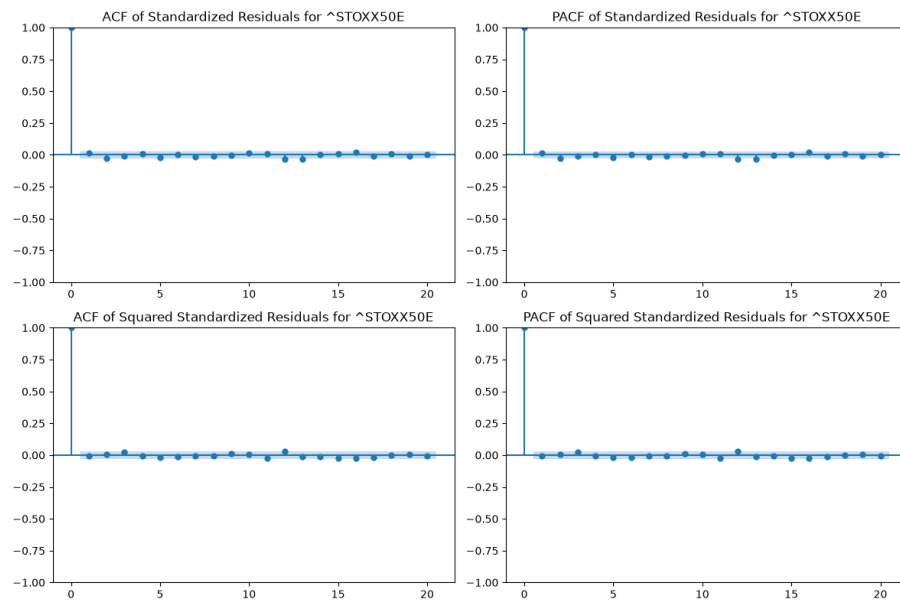
index	model	test	lag	stat	p_value
^FTSE	AR(1)- GARCH(1,1)- t	lb_std_resid 10		12.9257	0.2279
^FTSE	AR(1)- GARCH(1,1)- t	lb_std_resid 20		23.713	0.2552
^FTSE	AR(1)- GARCH(1,1)- t	lb_std_resid 10 squared		8.4607	0.5839
^FTSE	AR(1)- GARCH(1,1)- t	lb_std_resid 20 squared		17.7075	0.6067
^GSPC	AR(1)- GARCH(1,1)- t	lb_std_resid 10		14.275	0.1608
^GSPC	AR(1)- GARCH(1,1)- t	lb_std_resid 20		31.4188	0.0499
^GSPC	AR(1)- GARCH(1,1)- t	lb_std_resid 10 squared		5.6836	0.8411
^GSPC	AR(1)- GARCH(1,1)- t	lb_std_resid 20 squared		11.8408	0.9214
^N225	AR(1)- GARCH(1,1)- t	lb_std_resid 10		8.6317	0.5674
^N225	AR(1)- GARCH(1,1)- t	lb_std_resid 20		15.027	0.7749
^N225	AR(1)- GARCH(1,1)- t	lb_std_resid 10 squared		30.0278	0.0008477
^N225	AR(1)- GARCH(1,1)- t	lb_std_resid 20 squared		44.6328	0.0012
^STOXX50E	AR(1)- GARCH(1,1)- t	lb_std_resid 10		8.4025	0.5896
^STOXX50E	AR(1)- GARCH(1,1)- t	lb_std_resid 20		23.0496	0.2864

index	model	test	lag	stat	p_value
^STOXX50E	AR(1)- GARCH(1,1)- t	lb_std_resid_10	quared	5.9099	0.8228
^STOXX50E	AR(1)- GARCH(1,1)- t	lb_std_resid_20	quared	20.41	0.4326

The following ACF/PACF plots show the autocorrelation structure of the standardized residuals and squared standardized residuals from the fitted AR(1)-GARCH(1,1)-t models. As most autocorrelations lie within the confidence bands after lag 0, the models appear to fit relatively well.

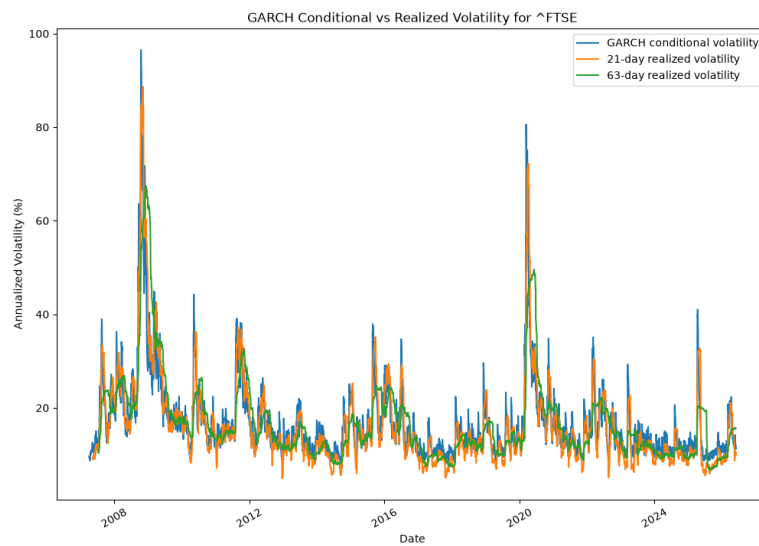


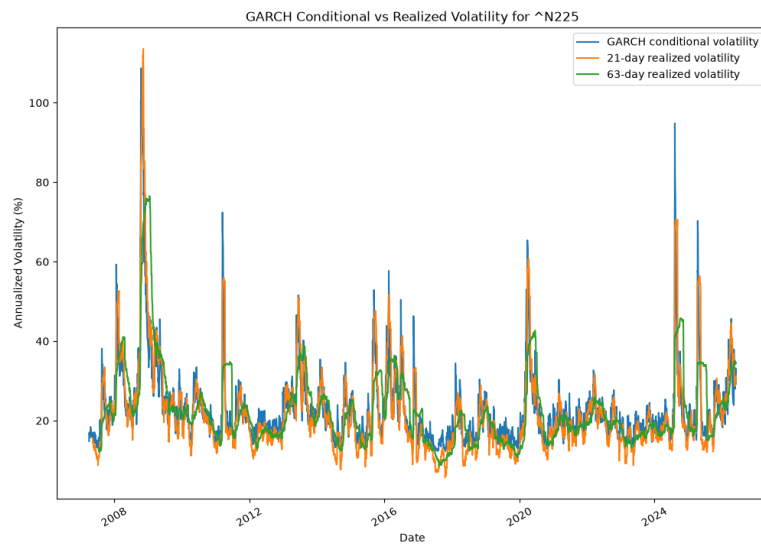
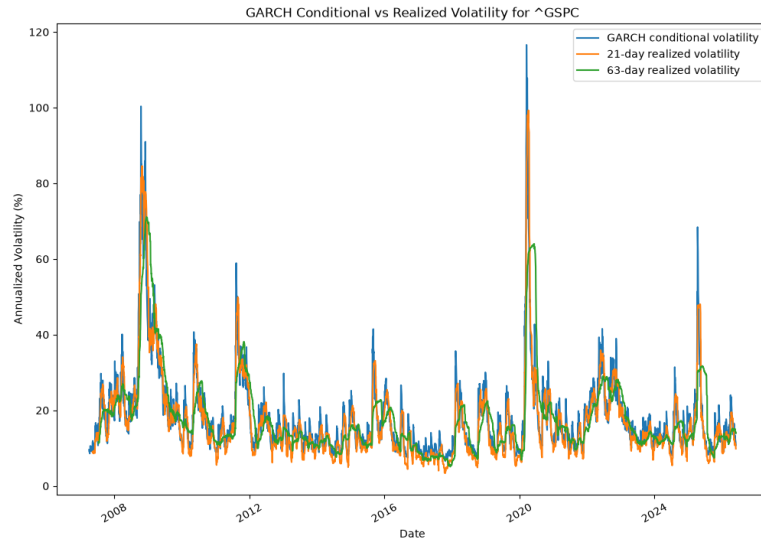


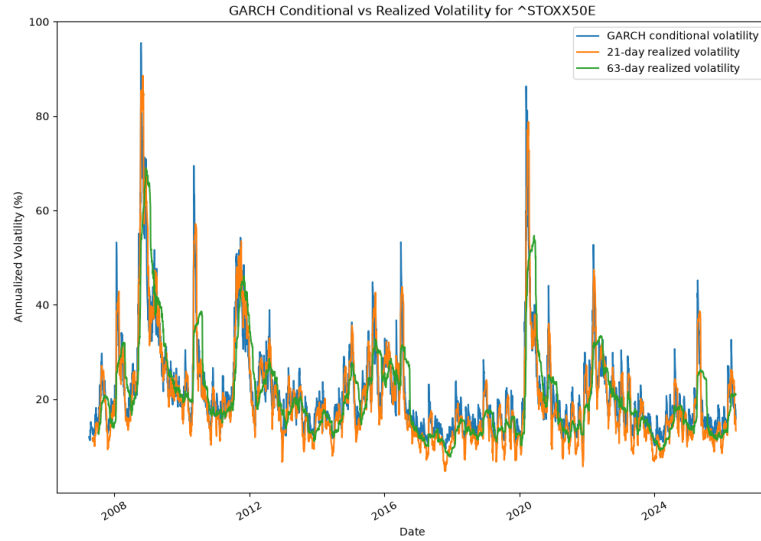


8. Volatility Forecasting

8.1 Conditional and Realized Volatility

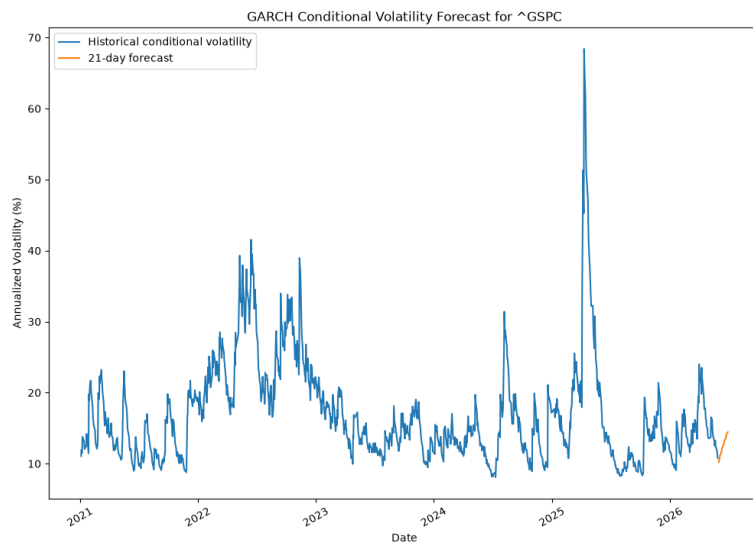
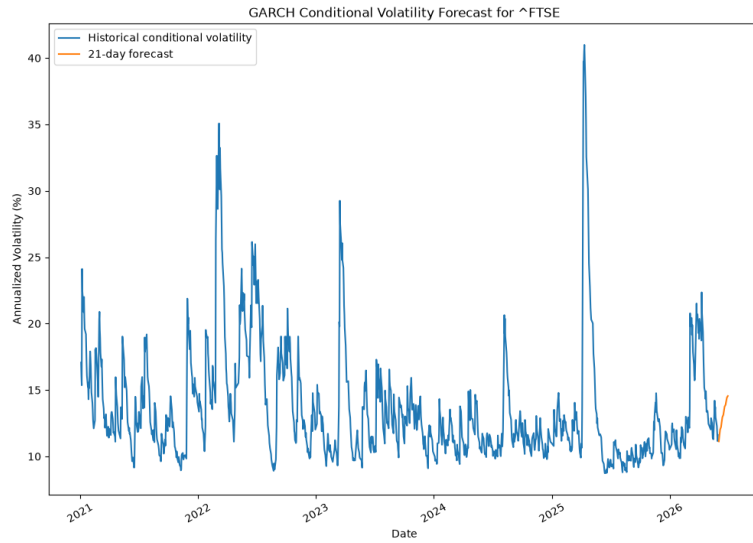


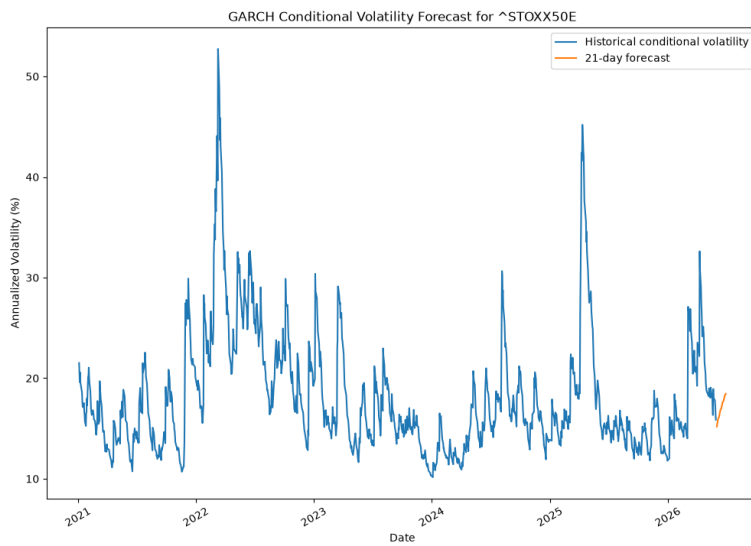
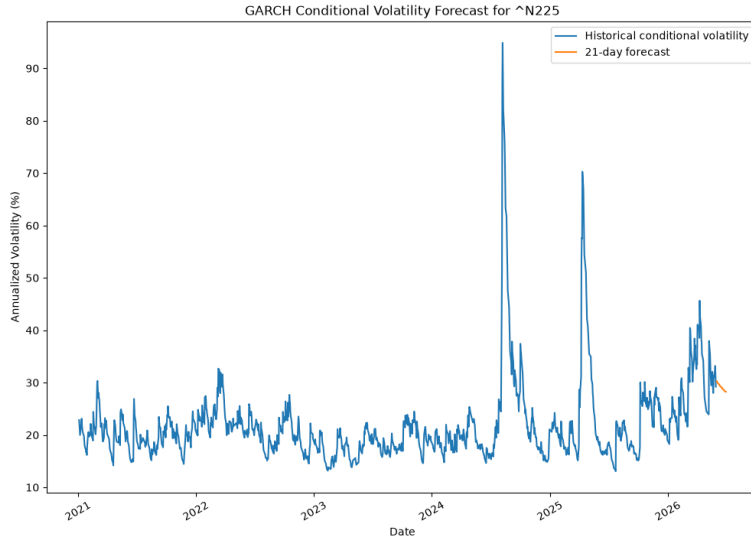




8.2 Forecasts

index	model	h	forecast_mean	forecast_variance	forecast_volatility	realized_forecast_volatility
^FTSE	AR(1)- GARCH(1,1)- t	21	0.0518	0.8398	0.9164	14.5478
^GSPC	AR(1)- GARCH(1,1)- t	21	0.1044	0.8293	0.9106	14.4561
^N225	AR(1)- GARCH(1,1)- t	21	0.0803	3.1675	1.7797	28.2525
^STOXX50E	AR(1)- GARCH(1,1)- t	21	0.0738	1.3499	1.1619	18.4439

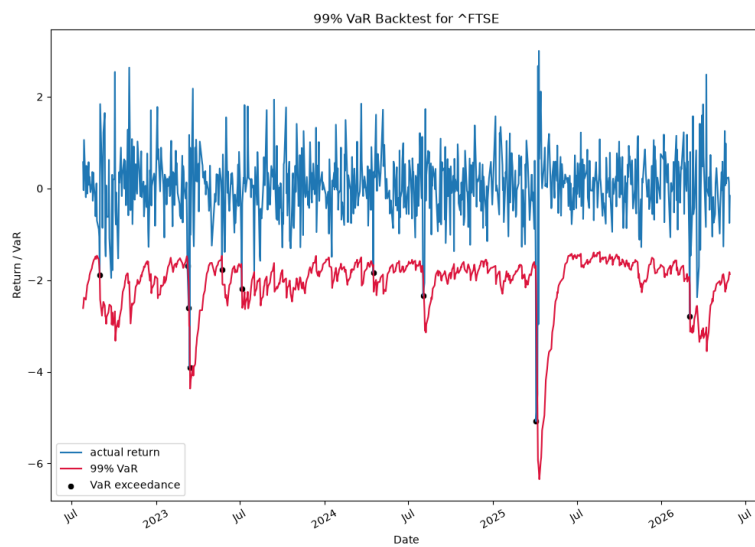


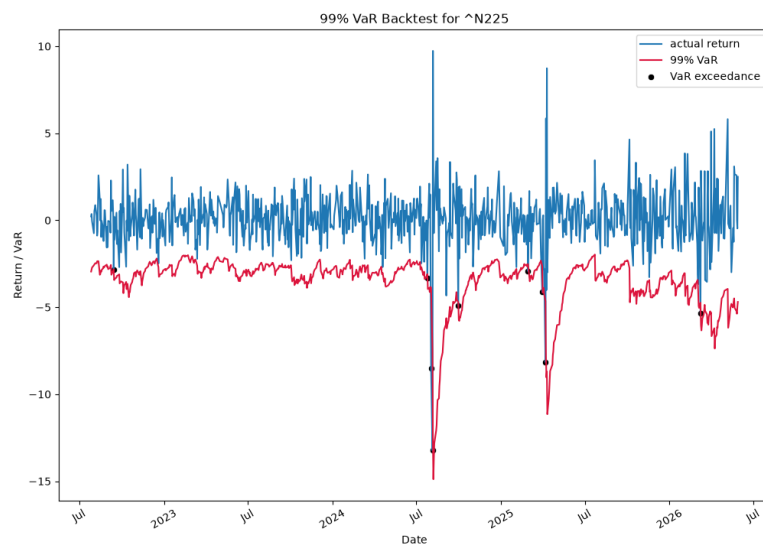
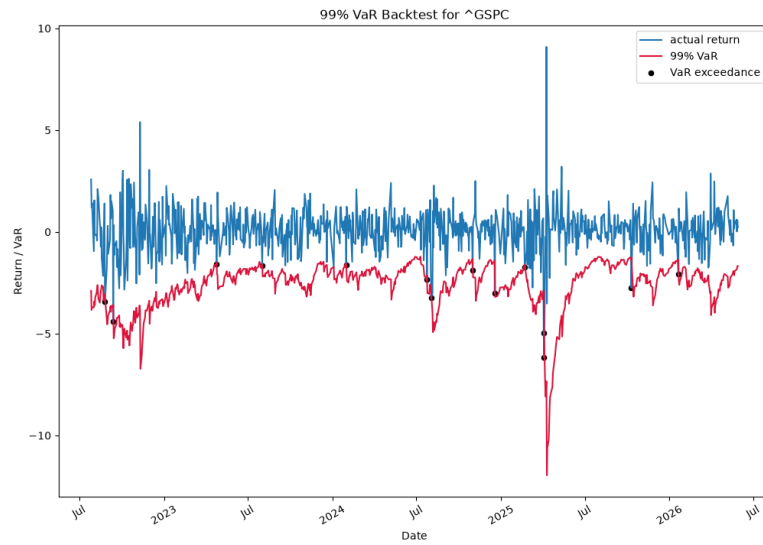


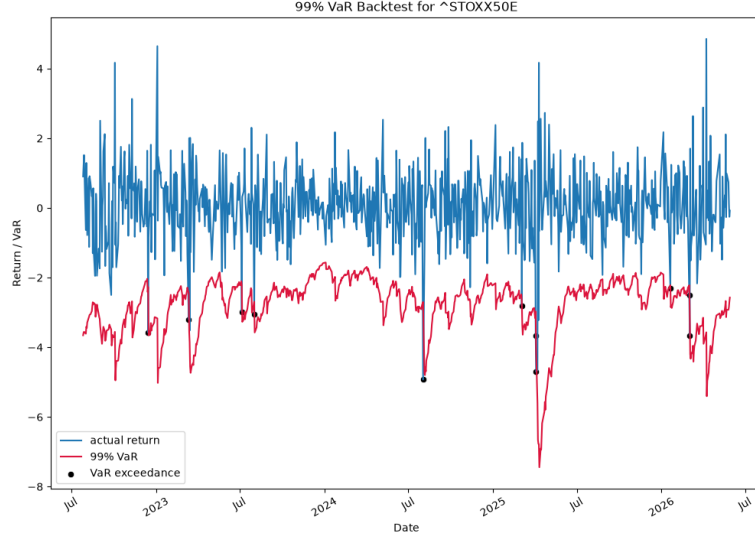
9. VaR Backtesting

The 99% VaR backtesting using an 80%-20% train-test data split shows that the model's VaR forecasts are broadly well-calibrated for all four indices. However, ^GSPC has the most exceedances and is the closest to potential underestimation of tail risk.

index	model	var_level	observations	exceedances	seedance	expected_exceedances	exceedance_rate	kupiec_pvalue
^FTSE	AR(1)- GARCH(1,1)- t	99%	875	10	8.75	0.0114	0.1724	0.678
^GSPC	AR(1)- GARCH(1,1)- t	99%	875	14	8.75	0.016	2.692	0.1009
^N225	AR(1)- GARCH(1,1)- t	99%	875	9	8.75	0.0103	0.0071	0.9326
^STOXX50E	AR(1)- GARCH(1,1)- t	99%	875	11	8.75	0.0126	0.5404	0.4623







10. Limitations

One limitation of the project is that the AR(1)-GARCH(1, 1) model does not fully remove the autocorrelation of the squared standardized residuals for $\hat{GSPC}$ and $\hat{N225}$. To address this, future projects could compare richer volatility models such as GJR-GARCH, EGARCH, or higher-order GARCH specifications. Another limitation is that each index is modeled separately, so the analysis does not capture cross-market dependence or spillover effects across global equity markets.